

## The zero-sum sheath stabiliser in JOREK

where it lives, what it looks like, and what you set — after SOLPS-ITER b2stbc\_stab\_coeff\_sheath\_te/ti/ni

**1. The input parameters.** Three reals in namelist in1, all default 0, which reproduces the previous code bit-for-bit:

```
stab_coeff_sheath_te = 100.d0 ! electron energy sheath row
stab_coeff_sheath_ti = 100.d0 ! ion energy sheath row
stab_coeff_sheath_ni = 100.d0 ! density sheath row
```

```
declared    models/phys_module.f90
defaults    models/preset_parameters.f90:81--83
namelist     models/model600/initialise_parameters.f90:67--68
broadcast    communication/broadcast_phys.f90 (packed/unpacked in matching order)
```

**2. The maths.** Let  $\Pi$  be the sheath *collection flux* — the geometric factor already multiplying every term in that boundary row:

$$\Pi = \underbrace{(\mathbf{B} \cdot \mathbf{n}) V_{\parallel} R \, dl}_{\text{parallel}} + \underbrace{v_{E,n}^{\text{clip}} R \, dl}_{E \times B} + \underbrace{c_s R \, dl \, c_{\alpha}}_{\text{grazing floor}}$$

The *physical* sheath energy source and the *added* stabiliser are

$$S_{T_i} = -(\gamma_{\text{sh},i} - 1) \rho T_i \Pi, \quad S_{T_i}^{\text{stab}} = +\alpha_i \rho \Pi (T_i^{\text{old}} - T_i^{\text{new}})$$

JOREK linearises about the old state, so with  $b$  the residual and  $\mathbf{A}$  the matrix:

$$S^{\text{stab}}|_{T^{\text{new}}=T^{\text{old}}} = 0 \Rightarrow \text{no residual contribution}, \quad \frac{\partial S^{\text{stab}}}{\partial T^{\text{new}}} = -\alpha_i \rho \Pi \Rightarrow A_{T_i T_i} += \theta \alpha_i \rho \Pi \hat{T}_i$$

so the assembled column becomes  $(\gamma_{\text{sh},i} - 1 + \alpha_i) \rho \hat{T}_i \Pi$  while the residual keeps  $(\gamma_{\text{sh},i} - 1)$ . **That asymmetry is the entire implementation.** The converged state is where  $b = 0$ , and  $\mathbf{A}$  does not enter that condition — so  $\alpha$  cannot move the answer.

**3. The code.** One file: models/model600/mod\_boundary\_matrix\_open.f90.

*The residual — untouched, keeps the physical  $\gamma$ :*

```
rhs_ij(var_Ti) = - v * (gamma_sheath_i-1.d0) * r0 * Ti0 * vpar0 * ps0_s * normal_sign3 * timestep &
- v * (gamma_sheath_i-1.d0) * r0 * Ti0 * fu_ven_sh * BigR * dl * timestep &
- v * (gamma_sheath_i-1.d0) * r0 * Ti0 * cs0 * BigR * dl * c_angle * timestep &
```

*The Jacobian — the added term (line 651). The bracket is  $\Pi$ : the same three factors as above.*

```
amat(var_Ti,var_Ti) = amat(var_Ti,var_Ti) &
+ v * stab_coeff_sheath_ti * r0 * Ti &
* ( vpar0 * ps0_s * normal_sign3 * ! parallel
+ fu_ven_sh * BigR * dl & ! clipped ExB
+ cs0 * BigR * dl * c_angle ) & ! grazing floor
* theta * timestep
```

$v$  = test function,  $Ti$  = *trial* function,  $r0/vpar0/cs0$  = lagged old-state values,  $\text{theta} \times \text{timestep}$  = the standard implicitness weighting used by every neighbouring entry.

**4. All four insertion points** — Jacobian diagonal only, nothing off-diagonal, no other equation:

| line | matrix entry          | parameter            | row               |
|------|-----------------------|----------------------|-------------------|
| 616  | amat(var_rho,var_rho) | stab_coeff_sheath_ni | density           |
| 651  | amat(var_Ti,var_Ti)   | stab_coeff_sheath_ti | ion energy        |
| 673  | amat(var_Te,var_Te)   | stab_coeff_sheath_te | electron energy   |
| 706  | amat(var_T,var_T)     | stab_coeff_sheath_ti | single- $T$ build |

**Careful:** had we scaled *both* residual and Jacobian we would simply have raised  $\gamma_{\text{sheath}}$  — a different sheath and a different steady state. Only the Jacobian is touched.

**5. What it does, in numbers.** Same equation  $(T^{\text{new}} - T^{\text{old}})/\Delta t = Q - kT + \alpha k(T^{\text{old}} - T^{\text{new}})$  with  $Q = 5$ ,  $k = 2$ ,  $\Delta t = 0.5$ , start  $T = 9$ , so  $T^* = Q/k = 2.5$ :

|                | start | 1     | 2     | 3     | 4     | 5     | 6     | after 4000 steps    |
|----------------|-------|-------|-------|-------|-------|-------|-------|---------------------|
| $\alpha = 0$   | 9.000 | 5.750 | 4.125 | 3.313 | 2.906 | 2.703 | 2.602 | <b>2.5000000000</b> |
| $\alpha = 100$ | 9.000 | 8.936 | 8.873 | 8.811 | 8.749 | 8.688 | 8.627 | <b>2.5000000000</b> |

Identical destination to ten digits;  $\alpha$  only takes smaller bites, so no single step can overshoot.

**6. Verification.** `test_sheath_stabiliser` in `tests/floating_transport/test_boundary.f90` assembles at  $\alpha = 0, 10, 20$  and asserts: residual **bit-identical**; Jacobian **exactly linear** in  $\alpha$ ; **nothing** outside the  $T_i/T_e/\rho$  diagonal blocks changes.

**7. Result.** Drift-enabled run: previously terminated at  $\leq 649$  steps  $\rightarrow$  **past 1100 and converging**.

Note: SOLPS iterates within a time step, so there the term vanishes before the step is accepted and is purely a convergence aid. JOREK takes one solve per step, so  $T^{\text{old}}$  is the previous *time step* and genuine transients are slowed as well. The stationary state is unaffected either way.